ACADEMIC CENTRAL

PRACTICE EXAMINATION

UNIVERSITY · 1ST YEAR

COMPUTER LITERACY

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 2 hours**

BSc Computer Science, 1st year, semester 2 theory examination. Section A: objective questions (25) · Section B: hardware and software, file management, networks, the internet and online safety, spreadsheets, word processing and presentations (75).

Written by Academic Central for practice, following first-year computer literacy (end-user computing) syllabuses at South African universities. It is not a university examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SEVEN questions. Answer ALL the questions.
2. Read each question carefully. Answer in full sentences where an explanation is asked for.
3. Write spreadsheet formulas exactly as you would type them in Microsoft Excel (starting with =).
4. Calculators are NOT allowed.
5. Number your answers exactly as the questions are numbered.

SECTION A

QUESTION 1: MULTIPLE CHOICE

Choose the correct answer. Write only the letter (A–D) next to the question number, e.g. 1.16 B.

1.1 Which of the following is an INPUT device?

A Monitor
B Printer
C Scanner
D Speaker

1.2 Which type of memory loses its contents when the computer is switched off?

A ROM
B RAM
C A solid state drive
D A flash drive

1.3 One byte is made up of …

A 4 bits.
B 8 bits.
C 16 bits.
D 1 024 bits.

1.4 Which component carries out the instructions of a program?

A The CPU
B The hard drive
C The monitor
D The power supply

1.5 Which of the following is SYSTEM software?

A Microsoft Word
B Windows 11
C Google Chrome
D Microsoft Excel

1.6 Which file is most likely a spreadsheet?

A budget.docx
B budget.xlsx
C budget.pptx
D budget.jpg

1.7 Which keyboard shortcut COPIES selected text in Windows?

A Ctrl + V
B Ctrl + X
C Ctrl + C
D Ctrl + Z

1.8 A network that connects the computers in one building or campus is a …

A LAN.
B WAN.
C URL.
D ISP.

1.9 Which protocol shows that a web page uses a secure, encrypted connection?

A HTTP
B HTTPS
C FTP
D SMTP

1.10 A fake email that tries to trick you into giving your password is an example of …

A spam.
B phishing.
C a firewall.
D a backup.

1.11 Cell A1 contains 3 and cell B1 contains 4. What does the formula =A1+B1*2 give?

A 14
B 11
C 10
D 24

1.12 In a spreadsheet, the cell reference A1 is …

A a relative reference.
B an absolute reference.
C a range.
D a function.

1.13 Which search technique finds pages with an exact phrase?

A Putting the phrase in quotation marks
B Typing the phrase in capital letters
C Adding a hashtag
D Using fewer words

1.14 Which South African law protects people's personal information?

A POPIA
B The Companies Act
C The Labour Relations Act
D The Consumer Protection Act

1.15 A backup copy of your work is safest when it is kept …

A on the same hard drive.
B on another device or in the cloud.
C in the Recycle Bin.
D in the Downloads folder.

[15]

QUESTION 2: TRUE OR FALSE

Write TRUE or FALSE next to the question number.

2.1 A solid state drive (SSD) has no moving parts.
2.2 Formatting a disk erases the data stored on it.
2.3 The operating system manages the computer's hardware and resources.
2.4 Bluetooth is a wired connection.
2.5 Your date of birth makes a strong password.
2.6 Cloud storage keeps files on remote servers that you reach over the internet.
2.7 In a word processor, a header appears at the bottom of every page.
2.8 Antivirus software protects you fully even if it is never updated.
2.9 A URL is the address of a web page.
2.10 Plagiarism is using someone else's work without acknowledging them.

[10]

SECTION B

QUESTION 3: HARDWARE AND SOFTWARE

3.1 Explain the difference between hardware and software. Give ONE example of each. (4)

3.2 Explain the difference between primary storage and secondary storage. Give ONE example of each. (4)

3.3 Name TWO functions of an operating system. (2)

3.4 Explain the difference between open-source software and proprietary software. (2)

3.5 A first-year student wants to buy a laptop. Name THREE specifications the student should compare, and say briefly why each one matters. (3)

[15]

QUESTION 4: FILE MANAGEMENT

Thandi saved her work at this location:
C:\Users\Thandi\Documents\CS1\Assignments\Assignment1.docx

4.1 Write down the file name and the file extension. (2)

4.2 In which folder is the file saved? (1)

4.3 What does the file extension tell us about the file? (1)

4.4 Explain the difference between Save and Save As. (2)

4.5 What is file compression (zipping)? Give ONE reason for using it. (2)

4.6 Give TWO guidelines for naming files sensibly. (2)

[10]

QUESTION 5: NETWORKS, THE INTERNET AND ONLINE SAFETY

5.1 Explain the difference between the internet and the World Wide Web. (2)

5.2 Give TWO advantages of connecting the computers in a computer lab in a network. (2)

5.3 What is the function of a router? (1)

5.4 A student's video lecture keeps stopping to buffer at home. Explain what bandwidth is, and how it causes this problem. (2)

5.5 Read the email below and answer the questions.

> **From:** ITS Helpdesk <its-support@univ-account-check.co>
> **Subject:** URGENT!!! Your student account will be closed today
>
> Dear Student,
> We detected a problem with you're account. To avoid losing access to your marks and emails, click the link below within 2 hours and enter your student number and password to verify your account:
> http://univ-login-verify.net/student
>
> ITS Helpdesk

5.5.1 Give THREE signs that this is a phishing email. (3)

5.5.2 What should the student do with this email? (1)

5.6 Give TWO features of a strong password. (2)

5.7 What is two-factor authentication (2FA)? (2)

5.8 What is malware? Name TWO types of malware. (3)

5.9 Give ONE responsibility that a university has under POPIA when it keeps students' personal information. (1)

5.10 Give ONE rule of netiquette for a class WhatsApp group or email to a lecturer. (1)

[20]

QUESTION 6: SPREADSHEETS

A lecturer uses the spreadsheet below. The final mark is 20% of Test 1, plus 20% of Test 2, plus 60% of the Exam.

	A	B	C	D	E	F
1	**Student**	**Test 1**	**Test 2**	**Exam**	**Final**	**Result**
2	Ayanda	65	72	58		
3	Bongani	40	35	48		
4	Chantelle	80	88	91		
5	Dumisani	55	49	52		
6	Esther	30	45	42		
7						
8	**Average**					

6.1 Write the formula for cell E2 that calculates Ayanda's final mark. (3)

6.2 Calculate Ayanda's final mark. Show your working. (3)

6.3 Write the formula for cell F2 that shows Pass if the final mark is 50 or more, and Fail otherwise. (3)

6.4 Write the formula for cell B8 that calculates the average Test 1 mark. (2)

6.5 Write a formula that counts how many students passed (use the Result column). (2)

6.6 Write a formula that finds the highest Exam mark. (1)

6.7 The lecturer types the weights 0.2, 0.2 and 0.6 in cells B10, C10 and D10. Rewrite the formula for E2 so that it uses these cells and can be copied down to E6 correctly. (3)

6.8 Which type of chart is best for comparing the final marks of the five students? Give a reason. (2)

6.9 Give the cell reference of Chantelle's Exam mark. (1)

[20]

QUESTION 7: WORD PROCESSING AND PRESENTATIONS

7.1 Name the word-processing feature that:

7.1.1 builds a contents page automatically from the headings in a document; (1)

7.1.2 shows the same text (for example the title) at the top of every page; (1)

7.1.3 starts a new page before the page is full; (1)

7.1.4 checks the document for spelling and grammar mistakes. (1)

7.2 Explain what a citation and a reference list are, and why students must use them in assignments. (2)

7.3 Give FOUR guidelines for designing good presentation slides. (4)

[10]

TOTAL: 100

END OF QUESTION PAPER